Marwadi University Marwadi Chandarana Group	**Marwadi University** **Faculty of Engineering &Technology**	
Subject: DSIP	**Aim: open ended problem-8**	
	Date:23-08-2026	**Enrollment No:92400133057**

Github Link:- https://github.com/manavkasundra/DSIP

INTRODUCTION TO IMAGE ENHANCEMENT

Overview of Image Enhancement:-

Image enhancement is simply the process of fine-tuning a digital picture so it looks clearer or highlights the details you actually care about. Instead of leaving a photo dull, blurry, or washed out, you adjust things like brightness, contrast, and sharpness to make key features stand out whether that's helping a doctor spot something on an X-ray or just making a personal photo pop.

Why Image Enhancement is Necessary:-

Images captured in real-world scenarios often suffer from various degradations:

1. Bad lighting – too dark to see details, or blown out and washed out.
2. Flat contrast – everything looks gray, objects disappear into the background
3. Noise – grainy speckles from high ISO or cheap sensors.
4. Hardware limits – sensors can only capture so much range between shadows and highlights.
5. Weather – fog, haze, smoke eating up clarity.

The Two Primary Ways to Enhance an Image:-

1. Spatial Domain Methods:-

Work directly on pixels.

- Point operations: Adjust each pixel based on its value alone (histograms, contrast stretching).
- Mask operations: Look at a pixel's neighbors to sharpen or smooth (filters, kernels).

2. Frequency Domain Methods:-

Transform to Fourier space first

- High-pass: Boost edges and detail (sharpening)
- Low-pass: Suppress noise and fine detail (smoothing)

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Histogram-Based Enhancement:-

Histograms map out how light and dark pixels are spread across a photo. Adjusting this distribution is a go-to spatial method because it is:

- **Fast and lightweight:** Runs quickly without heavy computational strain.
- **Simple to code:** Easy to build into standard processing pipelines.
- **Versatile:** Reliably improves all sorts of uneven lighting and faded scenes.
- **Hands-off:** Automatically balances out contrast without needing manual per-pixel tuning.

What is an Image Histogram?
An image histogram is a graphical representation of the tonal distribution in a digital image. It plots the number of pixels for each tonal value.

For an 8-bit grayscale image (0-255):-

- X-axis: Brightness levels from 0 (pure black) to 255 (pure white).
- Y-axis: How many pixels landed on each level.

Mathematical Representation:-
For an image with L intensity levels (typically L=256 for 8-bit images):

$$\mathbf{h(i) = n_i}$$

Where:

- h(i) = frequency of intensity level i
- n_i = number of pixels with intensity value i
- $i \in \{0, 1, 2, ..., L-1\}$

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Core Properties of an Image Histogram :-

A histogram acts like a diagnostic fingerprint for an image, revealing key details at a glance:

Tonal Spread: Shows how shades are dispersed across darks, midtones, and highlights.

Contrast Level: If the data clumps together in a narrow band, the image lacks contrast. If it stretches broadly across the scale, contrast is high.

Overall Brightness: The chart's center of mass tells you the lighting mood—skewing left means a dark image, while leaning right means a bright one.

Total Pixel Count: Adding up every single bar in the graph always equals the total resolution ($M \times N$ pixels) of the picture.

Normalized Histogram (Probability Density Function) :-

p(i) = h(i) / N

Where:-

- p(i) = probability of intensity level i
- N = total number of pixels in the image
- Sum of p(i) over all i = 1

Histogram Interpretation Examples:-

Dark Image: Pixels concentrated near 0
Bright Image: Pixels concentrated near 255
Low Contrast Image: Pixels concentrated in a narrow range
High Contrast Image: Pixels spread across the entire range

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CUMULATIVE DISTRIBUTION FUNCTION (CDF)

What Is a Cumulative Distribution Function (CDF)?

The Cumulative Distribution Function (CDF) represents the probability that a pixel will have an intensity value less than or equal to a certain value.

Mathematical Formulation:-

$$c(i) = \Sigma[_{j=0}{}_{i}]\ p(j)$$

Where:

- c(i) = cumulative probability up to intensity i
- p(j) = probability of intensity j

Key Properties of the CDF:-

Always climbs (Monotonic): Because probabilities are never negative, the curve only moves upward or stays flat—it never dips.

Strict Range (0 to 1): Tracks percentages, starting at baseline probability $c(0) = p(0)$ and topping out at exactly $c(L-1) = 1.0$ (100%).

Smooth & Stable: Because it accumulates data gradually, the CDF curve looks much smoother and less erratic than a spike-filled raw histogram.

CDF Visualization:-

A curve that starts at bottom-left (0,0) and rises to top-right (255,1).
Steep sections = lots of pixels at those intensities. Flat sections = dead zones.

Where CDF Is Used:-

Histogram Equalization: Acts as the mathematical blueprint to remap squeezed pixel values across the full dynamic range.

Histogram Matching (Specification): Inverted to force an image's lighting profile to match a specific target photo.

Image Segmentation: Helps locate ideal cutoff thresholds to separate foreground objects cleanly from their background.

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HISTOGRAM EQUALIZATION :-

What Is Histogram Equalization?

Histogram equalization is a contrast-boosting method that takes bunched-up pixel values and redistributes them evenly across the entire brightness scale. By spreading out the most crowded shades, it stretches the dynamic range so hidden details step forward.

The Core Concept:-

The goal is to design a mapping rule that takes the clumped, uneven histogram of the original image and flattens it out so every brightness level is represented as uniformly as possible.

Mathematical Derivation:-

Let:

- r = original intensity ($0 \leq r \leq 1$ for normalized images)
- s = new intensity after transformation
- T(r) = transformation function ($0 \leq T(r) \leq 1$)

What Equalization Actually Does for an Image:-

Boosts Contrast: Expands squished tones across the full spectrum from deep shadows to bright highlights.

Flattens the Profile: Produces a relatively balanced, even distribution across all available brightness bins.

Zero Guesswork: Runs entirely automatically based on the image data—no manual thresholds or tuning parameters required.

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HISTOGRAM MATCHING :-

What Is Histogram Matching?

Histogram matching (frequently called histogram specification) reshapes an image's tonal profile so its brightness distribution mimics a chosen target histogram—usually taken straight from a reference photo that already has the exact lighting, mood, or contrast you want.

Difference from Histogram Equalization:-

Aspect	Histogram Equalization	Histogram Matching
Target	Uniform histogram	Specified histogram
Process	Automatic	Requires reference
Output	Always flat histogram	Matches reference distribution
Flexibility	Fixed	Highly flexible

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Mathematical Framework:-

Given:-

- Source image with CDF: F_s(r)
- Target/reference image with CDF: F_t(z)

Steps:-

1. Compute CDF of source image: F_s(r)
2. Compute CDF of target image: F_t(z)
3. Find mapping: $z = F_t^{-1}(F_s(r))$

Discrete Implementation:-

For discrete images:-

1. Calculate source CDF: c_s(k)
2. Calculate reference CDF: c_t(j)
3. For each source intensity k, find j such that c_t(j) is closest to c_s(k)
4. Map $k \rightarrow j$

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Code:-

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
ref_img = cv2.imread('image1.png', cv2.IMREAD_GRAYSCALE)

def hist_match(src, ref):
   s_hist, _ = np.histogram(src, 256, [0, 256])
   r_hist, _ = np.histogram(ref, 256, [0, 256])
   s_cdf, r_cdf = s_hist.cumsum() / src.size, r_hist.cumsum() / ref.size
   return np.interp(s_cdf, r_cdf, range(256)).astype(np.uint8)[src]

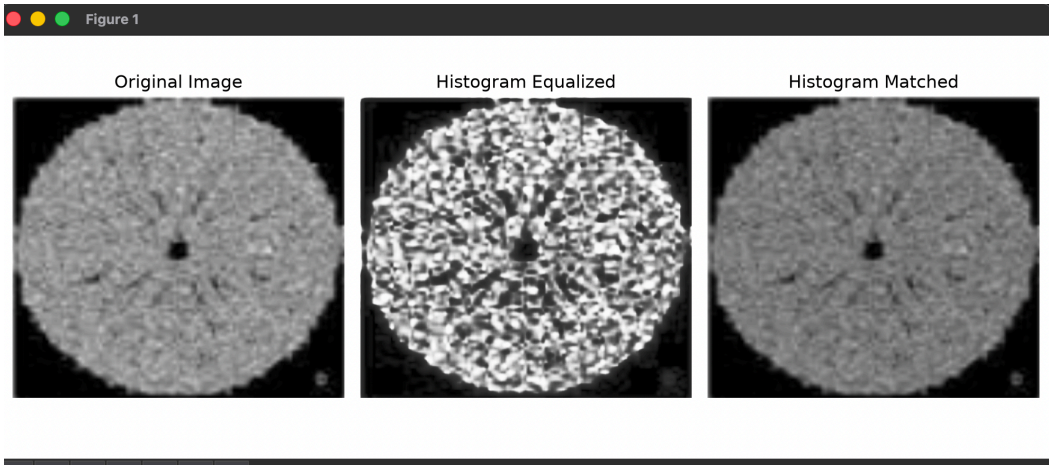
for i in range(1, 6):
   img = cv2.imread(f'image{i}.png', cv2.IMREAD_GRAYSCALE)
   if img is None: continue
   if img.dtype == np.uint16: img = (img / 256).astype(np.uint8)
   eq_img = cv2.equalizeHist(img)
   match_img = hist_match(img, ref_img)

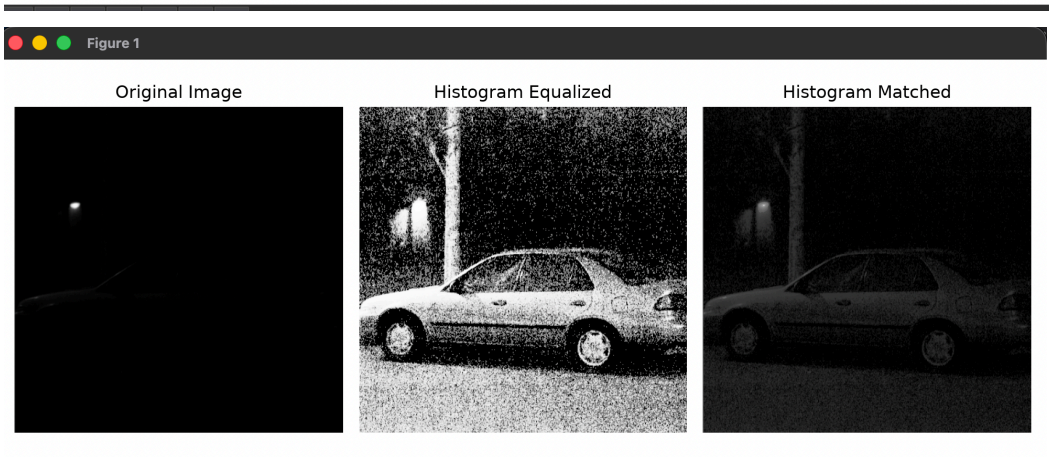
   eq_out = cv2.putText(eq_img.copy(), "Histogram Equalized", (10, 30),
cv2.FONT_HERSHEY_SIMPLEX, 0.7, 255, 2)
   match_out = cv2.putText(match_img.copy(), "Histogram Matched", (10, 30),
cv2.FONT_HERSHEY_SIMPLEX, 0.7, 255, 2)
   cv2.imwrite(f'image{i}_equalized.png', eq_out)
   cv2.imwrite(f'image{i}_matched.png', match_out)
   plt.figure(figsize=(10, 4))
   for idx, (title, im) in enumerate([('Original Image', img), ('Histogram
Equalized', eq_img), ('Histogram Matched', match_img)], 1):
       plt.subplot(1, 3, idx)
       plt.title(title)
       plt.imshow(im, cmap='gray')
       plt.axis('off')
   plt.tight_layout()
   plt.show()
```

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output:-

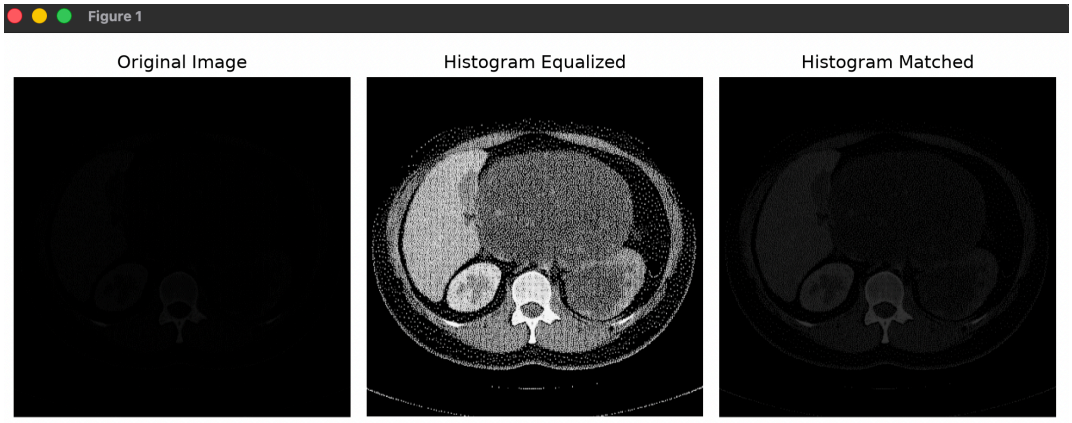






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Edge Cases:-

1. Constant Image:
   - Histogram has single peak
   - Equalization spreads values
   - Output may appear noisy
2. Saturated Image:
   - Many pixels at 0 or 255
   - Equalization compresses dynamic range
   - May lose detail
3. Small Images:
   - Histogram is sparse
   - Equalization may produce artifacts
   - More noticeable with fewer pixels

COMPARISON OF TECHNIQUES:-

Histogram Equalization vs. Histogram Matching

Aspect	Equalization	Matching
Goal	Uniform distribution	Match reference distribution
Parameters	None	Reference image
Automation	Fully automatic	Requires reference
Consistency	Same for all images	Consistent with reference
Use Case	General enhancement	Specific look/feel
Computational Cost	Lower	Slightly higher

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Visual Differences:-

Equalization:-

Pushes contrast to its upper limit.

Can end up looking overly processed or harsh.

Operates completely standalone without needing a second image.

Matching:-

Delivers a balanced, lifelike tone if the target image is well-shot.

Keeps the overall mood and global lighting intact.

Results hinge entirely on how good your reference shot is.

When to Choose Each Method:-

Choose Histogram Equalization when:-

1. Dealing with unpredictable or mixed lighting setups
2. You need a fast, hands-off enhancement
3. Speed and turnaround time are top priorities
4. No suitable target or reference image is available

Choose Histogram Matching when:-

1. You have an exact visual look or color profile in mind
2. Multiple shots must share an identical, cohesive style
3. Performing color palette transfers between photos
4. Standardizing tones across medical scans (X-rays, MRIs, CTs)

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Hybrid Techniques:-

Adaptive Histogram Equalization (AHE):-

1. Tweaks contrast block-by-block instead of treating the whole picture at once
2. Ideal for scenes with extreme shadow-to-highlight disparities
3. CLAHE is the most widely adopted version to keep noise in check

Adaptive Histogram Matching:-

1. Performs regional matching using local image patches
2. Preserves micro-textures and subtle fine details far better
3. Demands significantly more processing power and time

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Conclusion & Future Scope:-

The equalization and matching workflows built in this script offer dependable, automated ways to refine digital photos by rebalancing their tone distributions. Designed with a clean and modular structure in Python, the pipeline smoothly handles different bit depths and file formats, runs both transformation methods, and generates side-by-side plots for quick visual review. While standard equalization delivers an instant, hands-off contrast boost, matching grants precise creative and technical control by adopting the tonal profile of a reference photo. Together, these statistical approaches prove how practical histogram manipulation is for real-world tasks, setting up a solid base for advanced projects in clinical diagnostics, machine vision, and computational photography.

Histogram Equalization:-

1. Auto-balances brightness levels across the spectrum
2. Flattens the overall tonal profile
3. Stretches dynamic range with zero manual tuning

Histogram Matching:-

1. Aligns the input distribution with a chosen reference target
2. Offers predictable, stylized tonal adjustments
3. Relies on CDF curve comparison and linear interpolation

Implementation Highlights:-

1. Clean, modular Python and OpenCV workflow
2. Built-in support for different image types and depths
3. Side-by-side graphical visualizers for fast inspection
4. Safe error checks to skip missing inputs seamlessly